

Cloud Computing -- Guided Notes

Unit tp-2.11 | CompTIA Network+ N10-009 Obj. FC0-U71 2.11 | N10-008 FC0-U71 2.11

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Cloud computing delivers _____ resources (compute, storage, networking) over the Internet on demand. The three main service models are _____ (managed infrastructure -- VMs and storage), _____ (managed platform -- deploy apps without managing servers), and _____ (managed software -- use apps via browser).

Question 2

The four cloud deployment models are: _____ (resources owned and operated by one organization), _____ (resources provided by a vendor to the general public over the internet), _____ (two or more organizations share cloud infrastructure), and _____ (combines private + public cloud).

Question 3

_____ (the ability to grow resources up or out as demand increases) and _____ (the ability to shrink resources when demand decreases) are key cloud advantages. Combined, they allow organizations to pay only for what they _____ and avoid over-provisioning hardware that sits _____ during low usage periods.

Question 4

In cloud computing, the _____ model means customers pay only for what they use -- per hour, per GB, or per API call -- instead of large upfront hardware purchases. This converts capital expenditures (CapEx) to _____ expenditures (OpEx). The cloud provider owns and maintains the _____ infrastructure.

Question 5

A _____ is a geographic region where a cloud provider operates multiple data centers. Within a region, there are multiple _____ (physically separate buildings) to provide _____ -- if one building loses power, the others continue. Customers choose a region to reduce _____ for their users.

Question 6

SaaS (Software as a Service) examples include _____, _____, and _____ (list three examples). In SaaS, the provider manages the _____, _____, and _____ -- the customer just uses the application. The customer does _____ manage any servers or platform.

Question 7

A CDN (Content Delivery Network) stores _____ copies of content (images, videos, web pages) at servers geographically distributed around the world called _____ Points. When a user requests content, they are served from the _____ edge node, reducing latency and load on the _____ server.

Question 8 -- Real World Application

REAL WORLD: A startup is building a mobile app. They expect low traffic for the first 6 months, but if a major influencer promotes them they could see a 50x spike overnight. They have a budget of \$500/month at launch. Explain why the cloud is a better fit for this scenario than buying physical servers, and name the service model they would most likely use.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-2-11-cloud-computing/tp-2-11-cloud-computing-answer-key.pdf
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